

Geo-Location Intelligence & Advertisement Campaign Platform

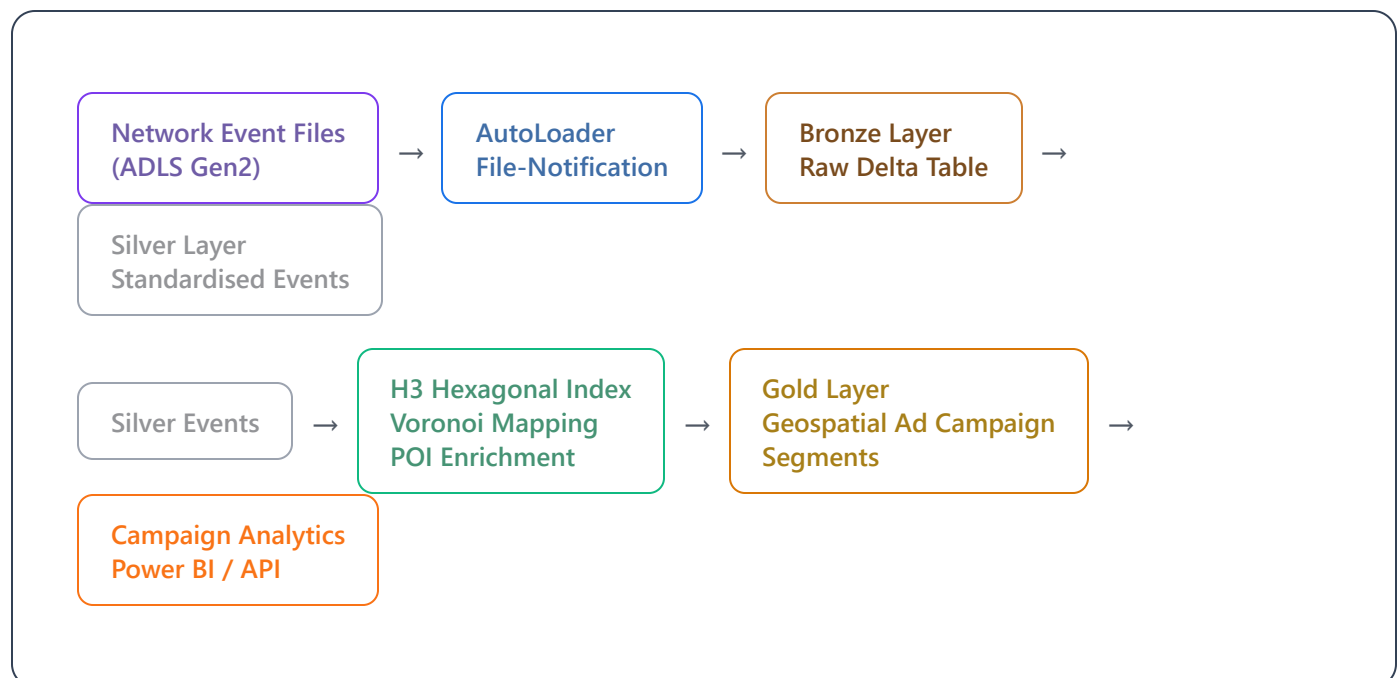
End-to-end geospatial analytics pipeline on Azure Databricks — from raw network event ingestion to location-based data powering advertisement campaign targeting and analytics.

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Solution Overview

This platform replaced a deprecated legacy geolocation system with a fully automated pipeline that ingests near-real-time network events via AutoLoader, enriches data with Uber H3 geospatial indexing, Voronoi/sub-Voronoi cell-site mapping, and POI reference data — maintained via SCD Type 2 monthly batch loads tracked at the Silver layer — then aggregates enriched results into Gold ad campaign segment tables. Parent-child notebook orchestration enables concurrent multi-campaign execution at scale.

High-Level Architecture



Data Layer Breakdown

INGESTION LAYER — AUTOLOADER (FILE NOTIFICATION MODE)

ADLS Gen2 Event Queue

Azure Event Grid

AutoLoader cloudFiles

Schema Inference

Checkpointing

Idempotent Writes

BRONZE — RAW DELTA TABLE

Raw network events

Ingest timestamp

File metadata

MD5 row hash

Partition by ingest_date

SILVER — STANDARDISED & VALIDATED EVENTS

Data type casting

Null / outlier filtering

Deduplication

Lat/Lon validation

Schema enforcement

POI Reference Data — Monthly Batch Load (SCD Type 2)

Monthly POI refresh

SCD Type 2 — poi_reference_history

MD5 change detection

MERGE INTO staging → history

eff_from / eff_to / crnt_flag

GEOSPATIAL PROCESSING (Processing Logic — not a Delta storage layer)

In-memory Spark transformations applied to Silver events. No intermediate Delta table is persisted — enriched output is written **directly to the Gold layer** tables.

Uber H3 Resolution 11

Voronoi Cell Sites

Sub-Voronoi Polygons

POI Lookup & Enrichment

OpenStreetMap Perimeter

Spatial Joins

GOLD — AD CAMPAIGN SEGMENTS

Enriched Geospatial Segments

Campaign Attribution Metrics

Partition by campaign_id

Delta OPTIMIZE scheduled

Analytics-ready schema

SERVING LAYER

Power BI Datasets

FastAPI / Databricks App

Campaign Compute Jobs

Control-M Scheduling

Orchestration Pattern

A parent notebook dynamically generates child notebook job submissions per campaign using `dbutils.notebook.run()` with concurrent threading (Python `ThreadPoolExecutor`). Each child processes one campaign segment independently, with configurable timeout, retry logic via **tenacity**, and granular status tracking written back to a control Delta table.

```
Parent Notebook
├── Load campaign config from Delta control table
├── ThreadPoolExecutor(max_workers=N)
│   ├── submit → child_notebook(campaign_id=001, timeout=3600)
│   ├── submit → child_notebook(campaign_id=002, timeout=3600)
│   └── submit → child_notebook(campaign_id=00N, timeout=3600)
└── Collect futures → write execution status to control table

Child Notebook
├── Read campaign parameters
├── Process and Write enriched output → Gold
└── Write status (SUCCESS / FAILED / PARTIAL) → control table
```

SCD Type 2 Implementation

Applied to the **Silver POI reference table** (monthly batch load). POI attributes — name, category, geometry, and type — are tracked over time. Each monthly refresh detects attribute changes via MD5 checksum and creates a new version, preserving full audit history with effective dates.

Column	Type	Purpose
record_key	STRING	Natural business key
row_hash	STRING (MD5)	Detect row-level changes
eff_from	TIMESTAMP	Record effective start
eff_to	TIMESTAMP	Record effective end (NULL = current)
crnt_flag	BOOLEAN	Identifies the current active record
inserted_at	TIMESTAMP	Audit: pipeline insert time

Spark Performance Optimisations

COMPUTE TUNING

- Adaptive Query Execution (AQE) — dynamic partition coalescing
- Photon vectorized engine for Delta scan & joins
- Optimise-write to minimize small file issues
- Auto-scaling job clusters

STORAGE TUNING

- Liquid Clustering on geo spatial join keys
- OPTIMIZE & VACUUM scheduled jobs
- Partition pruning on ingest_date

Technology Stack

Category	Technology
Compute	Azure Databricks (Photon clusters)
Storage	ADLS Gen2 + Delta Lake
Ingestion	Databricks AutoLoader (file notification)
Geospatial	Uber H3, Shapely, GeoPandas, OSMnx
Governance	Unity Catalog, Azure RBAC
Orchestration	Databricks Workflows + Control-M
Serving	Power BI, FastAPI, React JS
Alerting	SMTP, Databricks Notifications

Impact & Outcomes

- Replaced a manual legacy geolocation system with a fully automated, observable pipeline
- Enabled concurrent multi-campaign geo-computation with dynamic parallelism
- Enforced enterprise PII compliance via Unity Catalog fine-grained permissions
- End-to-end observability: Control-M scheduling, SMTP alerts, Power BI dashboards

- Unlocked incremental advertising revenue through location-based advertisement campaign targeting